

- DATA ANALYTICS PROJECT

When High Sales Hide Losses

How I uncovered ₹18K in silent losses inside a \$2.3M revenue business using SQL, Python & Power BI.

\$2.3M

Total Sales

12.47%

Profit Margin

10K

Orders Analysed

2014–17

Data Range

SQL · Window Functions

Python · Pandas · EDA

Power BI · DAX

Retail Analytics

Executive KPI Snapshot

2014–2017 consolidated performance

\$2.30M

TOTAL SALES



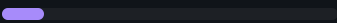
\$286K

TOTAL PROFIT



12.47%

PROFIT MARGIN



10K

TOTAL ORDERS



BUSINESS QUESTION

Is the business overall profitable, and where is there room to improve?

KEY FINDINGS

- \$2.3M revenue across 4 years with 10K orders — solid transaction volume.
- Business is profitable overall, but 12.47% margin leaves significant room for optimization.
- Category-level breakdown reveals the average masks serious losses in specific areas.

ANALYST INSIGHT

Headline metrics can be misleading. A 12% average margin hides sub-categories that are actively destroying value — exactly why granular analysis matters.

Sales vs Profit by Category

Does more sales always mean more profit?

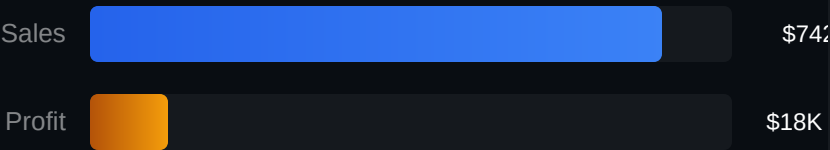
TECHNOLOGY



OFFICE SUPPLIES



FURNITURE ⚠



BUSINESS QUESTION

Which category generates the most revenue — and which actually makes the most money?

KEY FINDINGS

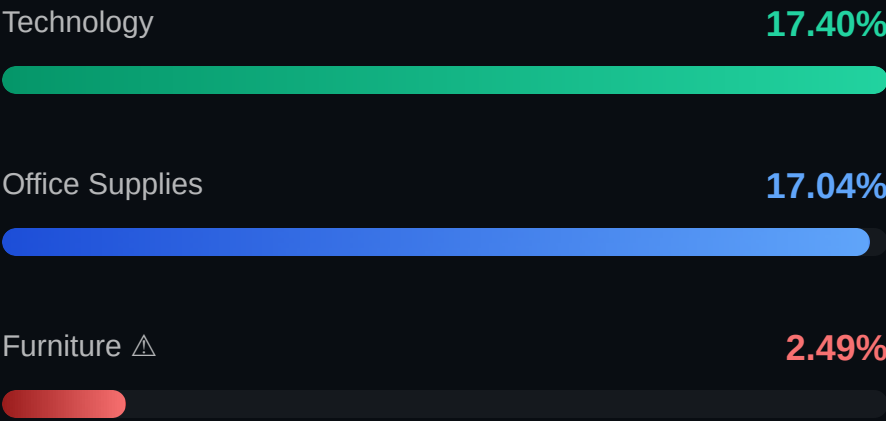
- Technology leads in both sales (\$836K) and profit (\$145K) — best-performing category.
- Office Supplies is consistent — strong margin relative to its sales volume.
- Furniture earns \$742K in sales but returns only \$18K profit — nearly invisible margin.

ANALYST INSIGHT

High revenue ≠ high profit. Furniture sells almost as much as Technology but captures 8× less profit. This is the central paradox this project was built to diagnose.

Profit Margin by Category

Efficiency of converting sales → profit



THE GAP

7× less efficient than Tech

BUSINESS QUESTION

Which category is most efficient at converting revenue into actual profit?

KEY FINDINGS

- Technology (17.4%) and Office Supplies (17%) are nearly identical — both highly efficient.
- Furniture's 2.49% margin is not just low — it's 7× less efficient than the best category.
- Furniture is not simply growing slower — it has a structural profitability problem.

ANALYST INSIGHT

The right response isn't to remove Furniture — it's to fix its discount policy and pricing strategy. Removing a \$742K revenue stream creates operational risk without guaranteed profit gain.

Sub-Category Profit Impact

Where exactly is value being destroyed?

SUB-CATEGORY	PROFIT IMPACT
Tables -8.6% margin	-\$18K
Bookcases -3.0% margin	-\$3K
Supplies -2.5% margin	-\$1K
Chairs +8.1% margin	+\$26K
Machines +1.8% margin	+\$3K

BUSINESS QUESTION

Which specific sub-categories are silently destroying profitability, and which are the hidden winners?

KEY FINDINGS

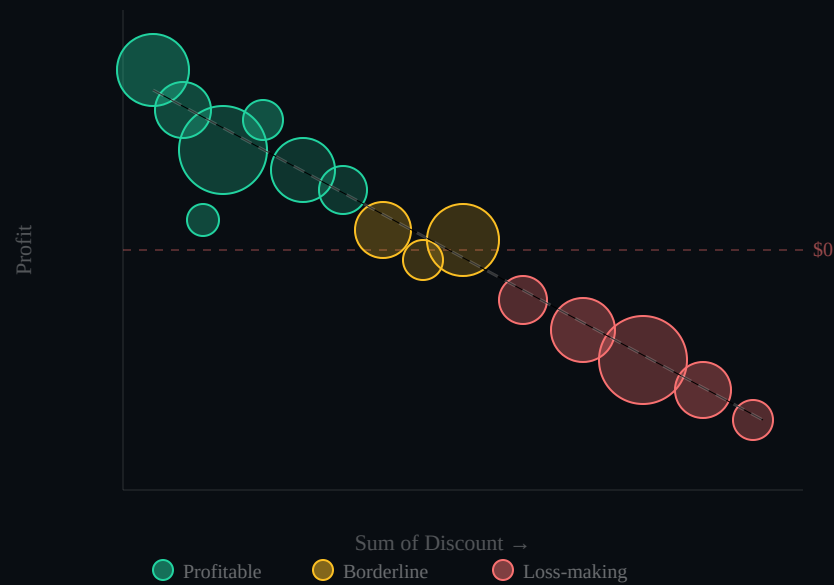
- Tables alone burn -\$18K — the single biggest profit destroyer in the entire dataset.
- Bookcases & Supplies add another -\$4K in losses — all within the Furniture category.
- Chairs earn +\$26K — proving Furniture itself isn't broken, just certain sub-categories.

ANALYST INSIGHT

The problem is surgical, not systemic. Chairs are profitable inside Furniture. The issue is Tables' pricing structure — reducing Tables' discount from aggressive levels could flip it from -\$18K to profitable.

Discount Impact on Profitability

The invisible margin killer



BUSINESS QUESTION

Does discounting grow the business, or is it silently eroding profitability?

KEY FINDINGS

- Clear negative correlation: as discount increases, profit consistently falls — often into negative territory.
- Moderate discounts (low range) still generate profit — controlled discounting has a role.
- Heavy discounters fall below the zero-profit line — the business pays customers to take products.

ANALYST INSIGHT

Discount is not inherently bad — it's uncontrolled discount that destroys margin. Each sub-category needs its own maximum discount threshold, calculated from its base gross margin.

Sales & Profit by Region

Which region converts sales most efficiently?



BUSINESS QUESTION

Which regions are most effective at converting sales into profit — and which are underperforming?

KEY FINDINGS

- West leads in both total sales (\$725K) and profit — highest efficiency region.
- East performs consistently well — second strongest profit-to-sales ratio.
- Central & South have disproportionately low profit vs their sales — discount or cost structure issue.

ANALYST INSIGHT

West's playbook — product mix, pricing discipline, or customer segment quality — should be analyzed and replicated in Central and South. Region is a lever, not just a label.

OPEN TO OPPORTUNITIES

Let's Turn Data into Decisions

I built this end-to-end analytics project — from raw CSV to boardroom-ready insights — using SQL, Python & Power BI. Looking for Data Analyst roles where I can create real business impact.

SQL · Window Functions · CTEs

Python · Pandas · EDA

Power BI · DAX · Power Query

Data Storytelling

Business Analytics

[View Full Project on GitHub →](#)

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